

Toda extensión que sea un módulo finitamente generado, es entera

Teorema 1. Sea B/A una extensión de anillos tal que B es un A -módulo finitamente generado, entonces B/A es una extensión entera.

Demostración: Sea $\{b_i\}_{i=1}^n$ un conjunto de generadores de B como A -módulo. Sea $c \in B$ no nulo, entonces $\forall i \in \mathbb{N}_n : c \cdot b_i \in B$, luego existen $a_{ij} \in A$ tales que

$$\left. \begin{array}{l} c \cdot b_1 = a_{11}b_1 + a_{12}b_2 + \cdots + a_{1n}b_n \\ c \cdot b_2 = a_{21}b_1 + a_{22}b_2 + \cdots + a_{2n}b_n \\ \vdots \\ c \cdot b_n = a_{n1}b_1 + a_{n2}b_2 + \cdots + a_{nn}b_n \end{array} \right\} \iff \left\{ \begin{array}{l} (a_{11} - c)b_1 + a_{12}b_2 + \cdots + a_{1n}b_n = 0 \\ a_{21}b_1 + (a_{22} - c)b_2 + \cdots + a_{2n}b_n = 0 \\ \vdots \\ a_{n1}b_1 + a_{n2}b_2 + \cdots + (a_{nn} - c)b_n = 0 \end{array} \right.$$

Podemos escribir este sistema en forma matricial como

$$\underbrace{\begin{pmatrix} a_{11} - c & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - c & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - c \end{pmatrix}}_{M_c} \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}}_b = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Entonces, $\text{Adj}(M_c^T) M_c b = 0 \implies \det(M_c) b = 0$. Entonces, $\forall i \in \mathbb{N}_n : \det(M_c) b_i = 0$.

Como los b_i generan B como A -módulo, existen $\{\alpha_i\}_{i=1}^n \subset A$ tales que

$$\alpha_1 b_1 + \alpha_2 b_2 + \cdots + \alpha_n b_n = 1 \implies (\det M_c) \left(\sum_{i=1}^n \alpha_i b_i \right) = \det(M_c) \implies \det(M_c) = 0.$$

Por tanto, $\det(M_c)$ es un polinomio mónico en c con coeficientes en A tal que $\det(M_c) = 0$, luego c es entero sobre A . ■

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.